

Only Destruction Survives: Honest Evaluation of Attribute Removal in Learned Representations

Anonymous submission

Abstract

Methods that remove a protected attribute from a learned representation are routinely certified with weak probes. We re-audit such certificates under a strong attacker battery and find that only 3 of 21 previously “stopped” verdicts survive. Across projection, adversarial scrubbing, and sealed-channel designs, erasure shaped against a probe fools only that probe. We characterize why: task outputs leak whatever information they actually use, so output-side leakage is bounded below by a floor set by the label–attribute coupling, which a one-feature test predicts in advance (Pearson $r = 0.795$; the predictor is an upper bound that is tight when the task is well learned). We propose a training procedure that concentrates attribute information into a designated subspace and destroys it, removing representation-side leakage at near-zero utility cost and leaving only the output-level floor, which we show is fundamental. Certification depends on attacker knowledge: against black-box attackers our method certifies at 85–102% retained utility, while informed attackers with clean-data side information are stopped only by full-rank noise at substantial cost. Under identical two-tier evaluation, 1 of 36 published-baseline combinations certifies, and the one survivor does so through incidental sampling noise rather than its fairness mechanism.

Introduction

Machine learning systems are increasingly required to make predictions without exposing protected attributes such as race or sex. A large body of work answers this requirement at the level of the representation: train or transform an encoder so that the attribute can no longer be recovered from it, then certify compliance by probing the representation and reporting that recovery is at chance. The certificate is only as strong as the probe. In practice the probe is almost always weak, a linear model or the method’s own adversary, and the guarantee inherits that weakness silently.

This paper asks what happens to such certificates under honest evaluation, and what honest protection actually costs. We assemble a strong attacker battery, gradient-boosted trees, deep probes, low-rank adaptation probes, and an informed likelihood-ratio attacker with clean-data side information,

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and re-audit compliance verdicts that weak certificates had granted. The result is stark. Of 21 verdicts in which a weak certificate reported an attribute “stopped,” 3 survive the battery. The failures are not close: representations certified at near-zero linear recoverability routinely yield the attribute to a tree-based probe at AUC above 0.9. Linear certificates, though deliberately scoped to linear adversaries (Belrose et al. 2023), leave a nonlinear gap that practical evaluations rarely check. We quantify that gap, and we find it is the rule rather than the exception.

The failures share a mechanism. Every method we audit that attempts to *hide* attribute information, orthogonal projections, kernel-statistic penalties, adversarial scrubbing, sealed architectural channels, fails against attackers it was not shaped against, because the information is still present. The only mechanism that survives is *destruction*: adding noise that removes the information outright. This is not an artifact of our implementations. Under identical two-tier evaluation, published baselines fail the same way on their authors’ own code, and the single published method that certifies anywhere does so through the sampling noise its variational objective happens to inject, not through its fairness machinery.

Honest destruction has a price, and the price obeys a law. Task outputs leak whatever information they actually use, so when a task genuinely depends on an attribute, the model’s decisions reveal it regardless of how thoroughly the representation is scrubbed. We measure this floor across tabular and vision cells and find it tracks the label–attribute coupling along the diagonal. The coupling is measurable in advance by a one-feature probe of the labels alone, giving a diagnostic that predicts the utility cost of honest removal before any training run (Pearson $r = 0.795$ across 20 natural cells, confirmed causally on constructed cells). The diagnostic is an upper bound on output leakage. It is tight when the task is well learned and loose otherwise, a boundary we establish by falsifying our own prediction on a vision cell and reconciling the miss.

Within the regime the law permits, removal can be made nearly free. We propose a training procedure that shapes the representation to concentrate attribute information into a designated subspace and then destroys that subspace with noise. Ablations show both halves are necessary: shaping without destruction leaks, destruction without shaping pays full-rank prices. Against black-box attackers the procedure

certifies at 102%, 100%, and $85 \pm 2\%$ retained utility on our three headline cells at a 0.55 recovery bar (90% at a relaxed 0.57 bar), removing representation-side leakage at near-zero cost and leaving only the output-level floor, which is fundamental. Certification is relative to attacker knowledge: an informed attacker that knows the channel and the clean-data statistics defeats every subspace-targeted defense by reading the noise-free complement, and is stopped only by full-rank noise at substantial utility cost. Protection, we find, must be priced by threat tier.

Our contributions:

- **An honest re-audit.** Under a strong two-tier attacker battery, 3 of 21 weak-certificate verdicts survive; probe-shaped erasure fools only the matching probe (Fig. 1).
- **Two propositions.** Linear certificates are provably invariant under linear downstream adaptation, so their apparent stability against linear attack is vacuous (Prop. 1); and no certificate computed from second-moment statistics can be non-vacuous at practical operating points (Prop. 2). Together they imply adversarial empirical evaluation is not optional.
- **A law and a diagnostic.** Output-side leakage is floored by label–attribute coupling (Fig. 2); a one-feature test predicts removal cost in advance, as an upper bound with a measured validity condition (Fig. 3).
- **A method reaching the floor.** Shape-then-destroy training removes representation-side leakage at near-zero cost at Tier 1 and characterizes the full price of Tier 2 (Fig. 5).
- **A provenance-audited gauntlet.** 1 of 36 published-baseline combinations certifies under identical evaluation, on official code where it exists (Table 1).

Related Work

Removing attributes from representations. Fair representation learning trains encoders so that protected attributes cannot be recovered downstream, by adversarial games (Madras et al. 2018; Ganin and Lempitsky 2015), variational objectives (Louizos et al. 2016), or closed-form projection (Belrose et al. 2023). Each certifies compliance against a stated adversary class, the method’s own adversary, a kernel statistic, or a linear probe. Linear certificates in particular are deliberately scoped: LEACE proves exact guardedness against linear adversaries and claims nothing beyond them (Belrose et al. 2023). Practical evaluations, however, rarely check beyond the certified class. We quantify the gap that habit leaves open, and find it is large enough to invert verdicts: representations certified compliant yield the attribute to a gradient-boosted probe at up to 0.99 AUC.

Evaluation as the weak link. That erasure does not transfer across adversary classes was established in kernel space by Ravfogel et al. (2022), and the privacy literature has long shown models leak attributes they were never trained to predict (Song and Shmatikov 2020). Probing methodology work makes the complementary point that what a probe finds depends on the probe (Voita and Titov 2020; Elazar et al. 2021). We build the graded version of this argument: a two-tier battery ordered by attacker knowledge, a systematic re-audit in

which 3 of 21 certified verdicts survive, and two propositions (Props. 1 and 2) showing the failure is not repairable within the certificate, since linear certificates are invariant to linear adaptation and second-moment certificates admit no non-vacuous bound.

Limits of removal. Information-theoretic tradeoffs between fairness and utility are classical (Zhao and Gordon 2019). Most directly, Stadler et al. (2024) formalize least-privilege learning and prove that representations cannot combine high task utility with the prevention of inference beyond the task label. Their theorem establishes that a leakage floor exists, at the level of representations and task-label leakage, which is slightly more general than our output-side phrasing. We contribute the empirical face of that limit: the floor tracks label–attribute coupling along the diagonal, a one-feature test predicts it before training, constructed cells confirm the relationship is causal, and our method approaches the floor rather than pretending to beat it.

Certified fairness. FARE (Jovanović et al. 2023) derives practical certificates that bound the unfairness of downstream classifiers drawn from restricted families. Our evaluation is the complement: empirical, adversarial, and graded by attacker knowledge, with Prop. 2 explaining why certificates computed from second-moment statistics cannot close the gap the restriction leaves open.

Adjacent evidence. The durability literature for language models reports the same pattern at larger scale: safeguards certified by their own evaluations fail under fine-tuning attacks (Tamirisa et al. 2025; Qi et al. 2025). Concurrent work on diffusion-model concept erasure likewise distinguishes guidance-based avoidance from destruction-based removal and finds that avoidance fools its own evaluation but not stronger probes (Lu et al. 2025). The mechanism our results isolate, that only information destruction survives, is the data-processing inequality read as an evaluation principle (Dwork and Roth 2014; Cover and Thomas 2006).

Setup: Cells, Battery, and Why Measurement Must Be Adversarial

We evaluate on three headline tabular cells, named by their label-coupling difficulty throughout: **easy** (HMDA, race $\rightarrow$ loan_decision, predictor 0.514), **middle** (HMDA, race $\rightarrow$ loan_amount_band, 0.584), and **hard** (Adult, sex $\rightarrow$ income, 0.603), plus two CelebA vision cells selected by the same statistic. Difficulty here is an output-side property, set by how strongly the label couples with the attribute, not a representation-side one: every clean representation we test leaks its attribute at essentially 1.0, so representation-side intuition orders these cells identically and wrongly. Attribute recovery is measured as macro one-vs-rest AUC, where 0.5 is chance and 0.55 is the protection bar. The recovery battery has two tiers, priced by attacker knowledge. Tier 1 is black box: XGBoost (Chen and Guestrin 2016), a deep MLP (256–256), and a rank-32 ReLU LoRA probe (Hu et al. 2022). Tier 2 adds a channel-aware Gaussian likelihood-ratio attacker that knows the clean representation and the noise law. The LoRA member is not redundant: in the end-to-end experiments it breached configurations that XGBoost and the

MLP both passed, so a battery without it would have certified operating points that do not hold. Certified operating points are the mean over three training seeds with the full battery on the representation and on the task outputs, and every headline point is re-run at five seeds. A method is only credited with hiding an attribute if the strongest probe we can build, at the stated knowledge tier, reads near chance on every exposed surface.

Proposition 1 (Linear certificates are blind to linear adaptation). *Let $h \in \mathbb{R}^d$ be a representation and z an attribute. For any invertible affine map A , the linear coefficient of determination satisfies $R_{\text{lin}}^2(Ah, z) = R_{\text{lin}}^2(h, z)$. Consequently a linear certificate cannot register any strictly linear downstream adaptation, and its stability under linear attack carries no evidential weight.*

Proposition 2 (Second-moment certificates are vacuous). *For certificates computed from second-moment statistics of a noise-protected representation, there exist data distributions with identical statistics under which an attacker achieves recovery approaching 1 at practical noise levels. No such certificate can be non-vacuous at the operating points used in practice.*

Certificates Lie and Hiding Fails

Fig. 1 is the audit of certificate-based erasure on the Adult cell. Every targeted projection, whether it minimizes class-conditional MMD in the style of LEOPARD (Saillenfest and Lemberger 2025) or an HSIC independence objective, drives the linear certificate to $R^2 \approx 0.036$, a clear pass at $\tau = 0.05$. The honest probe reads the same representations at 0.681–0.683 AUC, and the unprotected representation reads 0.683. The projections removed what the certificate can see and nothing that a gradient-boosted attacker uses. Only the noise channel moves the honest number, from 0.589 at $\sigma=1$ down to 0.514 at $\sigma=8$. On the HMDA cells the same projections never passed even the certificate (attacked R^2 plateaus at 0.166 with rank 32, against $\tau = 0.05$), which is its own indictment, so Adult is the cell where the certificate-pass contrast exists to be shown. Linear certificates, though deliberately scoped, leave a large nonlinear gap that practical evaluations rarely check. We quantify it: on this cell the gap is the full distance from a certified pass to unprotected-level recovery.

The Footprint Law and an A Priori Diagnostic

Hiding the representation is not the whole problem, because the task output is also an exposed surface. Fig. 2 plots, for each cell, the best-attacker AUC on the task outputs once the representation is hidden (the output-leak floor) against the label-coupling predictor, a single XGBoost fit recovering the attribute from the task label alone, computed before any model is trained. The six tabular cells track the diagonal, from (0.509, 0.509) to (0.676, 0.629). The causal law is that outputs leak what outputs actually use. The predictor is therefore an upper bound on the output floor, tight when the task is well learned and loose otherwise. The CelebA Attractive→Young cell makes the bound’s slack measurable: the label couples with Young at 0.734, but the deployable model only reaches

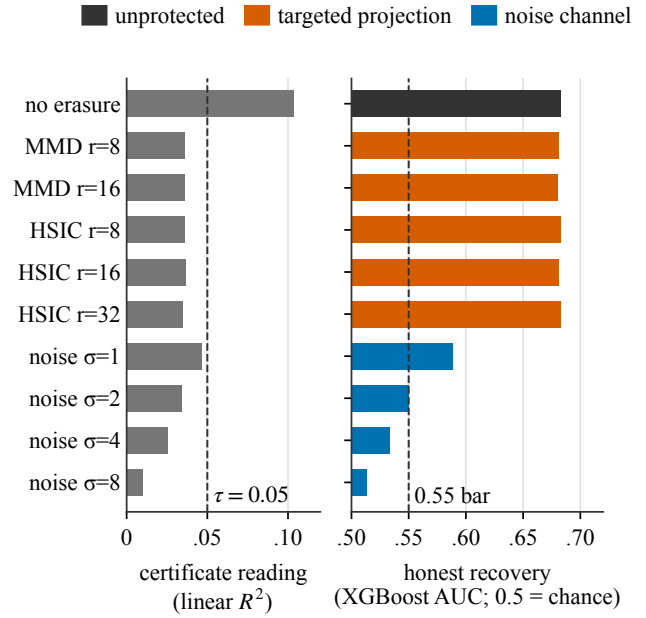


Figure 1: Certificate-based erasure audited on Adult (sex). Left: the linear certificate R^2 , where every projection passes at $\tau=0.05$. Right: honest recovery by XGBoost on the same representations, where every projection sits at the unprotected level (0.68, top bar) and only the noise channel descends toward chance. The certificate and the honest probe disagree about every projection and agree only about noise. On HMDA the projections never passed even the certificate, so no analogous contrast exists there.

0.565 accuracy, its achieved predictions couple with Young at 0.536–0.540, and the measured floor is 0.526–0.553, matching the achieved coupling rather than the label coupling. A weak model cannot betray label structure it never captured. Where removal is achievable the measured floors are small, 0.505–0.533 on the below-band cells (0.526–0.553 on the loose-bound CelebA cell), empirically approaching the leakage floor established by Stadler et al. (2024).

Fig. 3 turns the predictor into a forecast of cost. For 20 natural cells across three datasets, the durable-removal cost, the fraction of clean task lift lost at the noise level where the black-box battery first reads at or below 0.55 on both the representation and the output, correlates with the predictor at $r = 0.795$ and $\rho = 0.848$. Nine constructed cells, in which only the label–attribute coupling is tuned, reproduce the same rise, which makes the relationship causal rather than a dataset accident. The landscape is a steep ramp, not a cliff: inside the band 0.52–0.60 the cost climbs from near zero to near everything, and outside it the answer is essentially binary. In plain terms, one number computed from the labels alone, before training anything, tells a practitioner whether durable removal will be cheap, expensive, or hopeless.

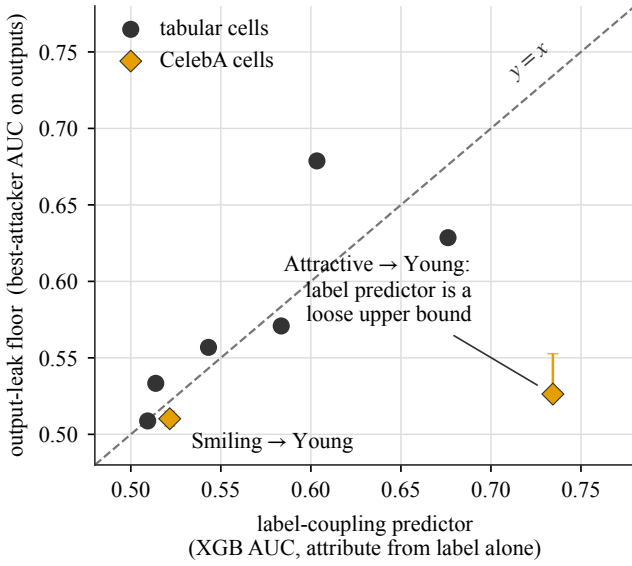


Figure 2: The output-leak floor against the label-coupling predictor, one point per cell, with $y=x$. Tabular cells track the diagonal: once the representation is hidden, the task output still betrays the attribute to the extent the label couples with it. The CelebA cells show the bound’s character: the predictor is an upper bound on the floor, tight when the task is well learned (Smiling→Young, on the diagonal) and loose when it is not (Attractive→Young, far below, whisker spanning the floors of the four erasure arms). There the achieved predictions couple with the attribute at 0.536–0.540 and the floor matches that value, not the label coupling of 0.734.

Reaching the Floor: Shape, Then Destroy

We propose to train the encoder and task head jointly with a shaping objective that concentrates attribute information into a designated subspace S , and to destroy S with noise before anything downstream can read it (Fig. 4). Both halves are necessary. Shaping without destruction is the projection failure of Fig. 1: the information survives whatever the shaping statistic cannot see. Destruction without shaping fails in both of its forms: noising an unshaped subspace leaves the attribute readable (XGBoost 0.84–0.87 and MLP 0.99 across the three cells at $\lambda=0$), and full-rank noise protects but pays the blunt price. The data-processing inequality is the qualitative reason destruction is attacker-agnostic, and by Prop. 2 it comes with no usable numeric second-moment ceiling, so the guarantee is, and remains, empirical.

Fig. 5 prices certified protection at both tiers for the end-to-end noised channel. Tier 1 removes representation-side leakage at near-zero cost on the achievable cells, leaving only the output-level floor, which is fundamental: the easy cell keeps $94.6\% \pm 2.2$ of its clean lift at $\sigma=8$ and the middle cell keeps $56.6\% \pm 0.7$ at $\sigma=12$. Tier 2 is strictly more expensive because the informed attacker must also be defeated: the same cells certify at $\sigma=20$ and $\sigma=24$ for $55.5\% \pm 4.4$ and $21.7\% \pm 0.5$. On the hard cell the diagnostic already predicts that useful certification is out of reach, and the certified points confirm it at -0.5% and -7.0% . Protection against a

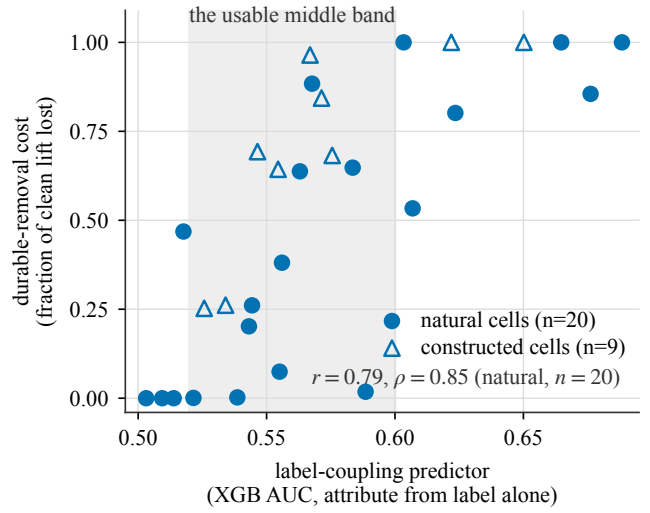


Figure 3: Durable-removal cost against the label-coupling predictor. Filled circles are 20 natural cells ($r=0.79$, $\rho=0.85$), open triangles are 9 constructed cells in which only the label–attribute coupling is tuned, so the rise is causal. The shaded band marks the usable middle (0.52–0.60), where the cost ramps from near zero to near total. Below the band removal is nearly free, above it removal costs essentially all task utility.

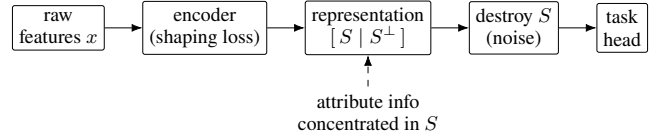


Figure 4: Shape-then-destroy. Training concentrates attribute information into subspace S ; S is destroyed with noise before the task head. Ablations show both halves are necessary.

stronger adversary is bought with more noise, and the price is read directly off the utility axis.

Surgical destruction, which confines the noise to the shaped attribute-carrying subspace, is a Tier-1 technology. It holds the black-box bar everywhere, but the informed likelihood-ratio attacker reads its untouched complement at 0.66–0.94, and no in-family escalation fixes that. Only full-rank noise buys Tier 2. Within Tier 1 the surgical channel is what makes the hard cell usable at all: it keeps $85\% \pm 2$ at the 0.55 bar (90% at a relaxed 0.57 bar), where the 90% point is the three-seed winner whose worst seed reads 0.561 and breaches at five seeds. One scope note: the end-to-end advantage over post-hoc surgery (104%, 100%, and 90% kept versus 88%, 81%, and 53% at matched Tier-1 protection, with the hard-cell 90% subject to the five-seed revision above) appears when representation training is under our control. On the frozen, pre-shaped CelebA representation the two tie (99.1% versus 98.4% on the easy vision cell).

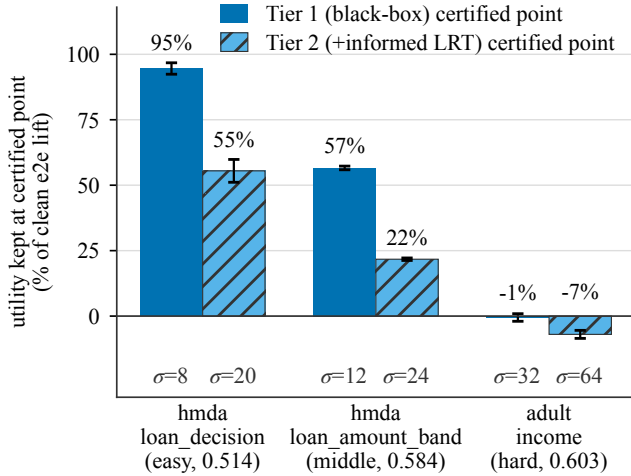


Figure 5: Utility kept at the certified operating point of the end-to-end noised channel, per cell and per tier (error bars: spread over three training seeds). Tier 1 (black box) is nearly free where the diagnostic predicts removal is achievable, and what remains exposed afterward is only the output-level floor, which is fundamental. Tier 2 (adding the informed likelihood-ratio attacker) is bought only by more full-rank noise, at $\sigma=20/24/64$, and its price is the visible drop. On the hard cell the diagnostic predicts no useful certified point exists, and none does.

The Gauntlet: Published Methods Under Identical Evaluation

We ran the field’s published fair-representation and erasure methods through the identical two-tier battery on the three headline cells, with knob sweeps, best honest operating point per tier, and three-seed certification. Each row carries a provenance tag (official code, validated reimplementation, or generic mechanism), summarized in Table 1 with per-baseline detail in the appendix. One of 36 method–cell–tier combinations certifies.

VFAE (Louizos et al. 2016) is the one survivor, and it certifies through its accidental sampling noise rather than its fairness machinery. The sampled- z exposure holds Tier 1 on the easy cell at 0.535 with 72% utility. The posterior-mean exposure of the same models reads 1.00 at every β , and raising the fairness knob makes sampled- z recovery strictly worse, 0.536, 0.583, 0.661, 0.743 for $\beta = 1, 10, 10^2, 10^3$, because a larger MMD weight shrinks the posterior noise faster than it matches groups. Its Tier-2 reading of 0.551–0.556 is at the bar, not under it, and with no noise knob it fails the middle and hard cells at 0.585 and 0.587. It is the exception that proves the mechanism: the only baseline that certifies anything does so exactly where the diagnostic says removal is cheap, via an unadjustable noise channel it does not control.

Oblivator (Akbari, Afshari, and Boddeti 2025), run from the authors’ pinned official code in its strongest supervised mode, fails Tier 1 on all three cells at every one of its 15 iterations, reading 0.968–0.999 while keeping 85–99% utility,

Method	Provenance	Utility	T1	T2
LAFTR	official+calib.	–	0.846+ ³	fail
VFAE	valid. reimpl.	72%	pass ¹	at bar
DANN-scrub	generic mech.	$\leq 58\%$ lost	fail	fail
LEACE	official pkg.	–	0.995+	fail
Oblivator	official (N’25)	–	0.97+	fail
Ours (post-hoc)	this work	53–89%	pass	fail
Ours (e2e)	this work	85–102%	pass	pass ²

Table 1: Gauntlet summary across the three headline cells and two threat tiers; 1 of 36 published-baseline combinations certifies. ¹Easy cell only, via incidental sampling noise. ²Tier-2 certification via full-rank noise at the costs of Fig. 5. ³Official-run range 0.846–1.000 (the 0.846 is one point, hard cell at $\gamma=8$); the validated multiclass reimplementation reads 0.995–1.000.

and its own MLP stopping probe reads 0.94–0.99 throughout, so its termination criterion never fires. This matches the prediction of our controlled HSIC projections (Fig. 1): on representations where the attribute is robustly and non-linearly encoded, HSIC-guided erasure has almost nothing kernel-visible left to remove, however seriously it is engineered.

LAFTR (Madras et al. 2018), both as our validated multiclass reimplementation and re-run on the official Vector Institute release under a compatibility shim calibrated to the paper’s own Adult numbers, fails Tier 1 everywhere, at 0.846–1.000 across every adversary weight. The official run adds one diagnosis: at $\gamma \geq 2$ on the easy cell the min-max “wins” fairness by zeroing task utility, lift falls to 3–17% of clean, while the representation still leaks at 0.996–1.000. **DANN-style adversarial scrubbing** (Ganin and Lempitsky 2015), our generic-mechanism baseline at full strength, genuinely removes signal but plateaus at 0.610–0.664 for 42–69% utility. **LEACE** (Belrose et al. 2023), from the authors’ library, reads 0.995–0.998 at nearly full utility, the unprotected level, as its linear scope predicts on representations that encode the attribute nonlinearly. In plain terms, no published method reaches the purpose-built channel on any cell at either tier, and the single certifying combination is an accident of a stochastic encoder.

Discussion

The floor is label-relative. The footprint law bounds leakage by what the task label couples to, and the label is a modeling choice. Our constructed cells show this operationally: tuning the label–attribute coupling down moves the floor down, with everything else fixed. A measured floor therefore mixes two things, genuine task necessity and bias inherited from the label itself. A lending label built from historical decisions may couple to race because the decisions did. When the diagnostic reads high, the first question is not how to remove harder but whether the label is the right one, since a cleaner label shrinks the wall that no removal method can.

Limitations. Our evidence is tabular plus one vision check, and the vision cells used a frozen, pre-shaped encoder,

which is exactly the regime where the end-to-end advantage disappears. The diagnostic is an upper bound whose tightness requires the task to be well learned. The 0.55 bar and the two tiers are choices, defensible but not canonical, and Tier-2 certification is expensive enough that for many deployments it amounts to a proof of infeasibility. We report certified operating points, not guarantees: by Prop. 2, no second-moment certificate can replace adversarial measurement, including ours.

Future work. The pattern we characterize, removal succeeding exactly where the target is weakly coupled to what the system must retain, suggests that for capabilities deeply entangled with model utility, preventing acquisition during training may dominate post-hoc removal. Testing this in language models, where fine-tuning attacks routinely undo safeguards (Tamirisa et al. 2025; Qi et al. 2025), is the natural next step. When the output must carry the attribute, output-side mechanisms such as differentially private responses (Dwork and Roth 2014) are the complementary direction.

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